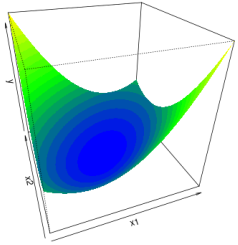
Optimization in Machine Learning

Mathematical Concepts
Quadratic functions I



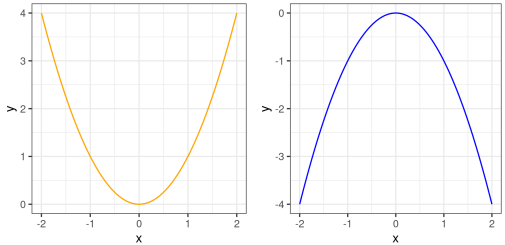
Learning goals

- Definition of quadratic forms
- Gradient, Hessian
- Optima

UNIVARIATE QUADRATIC FUNCTIONS

Consider a **quadratic function** $q : \mathbb{R} \to \mathbb{R}$

$$q(x) = a \cdot x^2 + b \cdot x + c, \qquad a \neq 0.$$



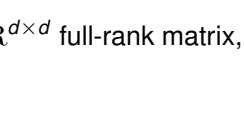
A quadratic function $q_1(x) = x^2$ (**left**) and $q_2(x) = -x^2$ (**right**).

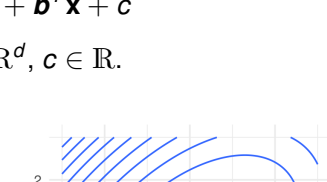
MULTIVARIATE QUADRATIC FUNCTIONS

A quadratic function $q : \mathbb{R}^d \to \mathbb{R}$ has the following form:

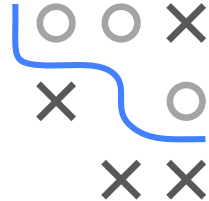
$$q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \boldsymbol{b}^T \mathbf{x} + c$$

with $\boldsymbol{A} \in \mathbb{R}^{d \times d}$ full-rank matrix, $\boldsymbol{b} \in \mathbb{R}^d$, $c \in \mathbb{R}$.





GRADIENT AND HESSIAN



- The **gradient** of q is

$$\nabla q(\mathbf{x}) = \left(\left(\mathbf{A}^T + \mathbf{A}\right)\mathbf{x} + \boldsymbol{b}\right)^T = \left(2\mathbf{A}\mathbf{x} + \boldsymbol{b}\right)^T \in \mathbb{R}^{1\times d}$$

  Derivative in direction $\boldsymbol{v} \in \mathbb{R}^d$ is (by chain rule)

$$\left.\frac{\mathrm{d}q(\mathbf{x} + h\cdot\boldsymbol{v})}{\mathrm{d}h}\right|_{h=0} = \left.\nabla q(\mathbf{x} + h\boldsymbol{v})\boldsymbol{v}\right|_{h=0} = \nabla q(\mathbf{x})\boldsymbol{v}.$$

- The **Hessian** of q is

$$\nabla^2 q(\mathbf{x}) = \left(\boldsymbol{A}^T + \boldsymbol{A}\right) = 2\mathbf{A} =: \mathbf{H} \in \mathbb{R}^{d\times d}$$

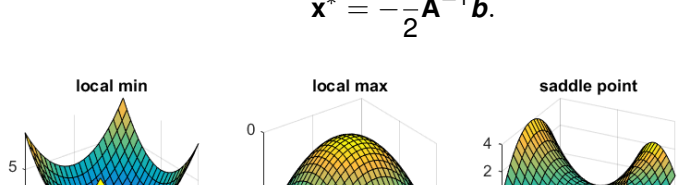
  Curvature in direction of $\boldsymbol{v} \in \mathbb{R}^d$ is (by chain rule)

$$\left.\frac{\mathrm{d}^2 q(\mathbf{x} + h\cdot\boldsymbol{v})}{\mathrm{d}h^2}\right|_{h=0} = \left.\boldsymbol{v}^T\nabla^2 q(\mathbf{x} + h\boldsymbol{v})\boldsymbol{v}\right|_{h=0} = \boldsymbol{v}^T\mathbf{H}\boldsymbol{v}.$$

OPTIMUM

Since **A** has full rank, there exists a *unique* stationary point $\mathbf{x}^*$ (minimum, maximum, or saddle point):

$$\nabla q(\mathbf{x}^*) = \mathbf{0}^T$$
$$(2\mathbf{A}\mathbf{x}^* + \boldsymbol{b})^\top = \mathbf{0}^T$$
$$\mathbf{x}^* = -\frac{1}{2}\mathbf{A}^{-1}\boldsymbol{b}.$$

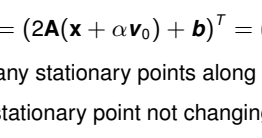


Left: A positive definite. **Middle: A** negative definite. **Right: A** indefinite.

OPTIMA: RANK-DEFICIENT CASE

Example: Assume $\mathbf{A}$ is **not** full rank but has a zero eigenvalue with eigenvector $\boldsymbol{v}_0$.

- Recall: $\boldsymbol{v}_0$ spans null space of $\mathbf{A}$, i.e., $\mathbf{A}(\alpha \boldsymbol{v}_0) = 0$ for each $\alpha \in \mathbb{R}$
- $\implies \mathbf{A}(\mathbf{x} + \alpha \boldsymbol{v}_0) = \mathbf{A}\mathbf{x}$
- Since $\nabla q(\mathbf{x}) = (2\mathbf{A}\mathbf{x} + \boldsymbol{b})^T$:

$$\nabla q(\mathbf{x} + \alpha \boldsymbol{v}_0) = (2\mathbf{A}(\mathbf{x} + \alpha \boldsymbol{v}_0) + \boldsymbol{b})^T = (2\mathbf{A}\mathbf{x} + \boldsymbol{b})^T = \nabla q(\mathbf{x})$$

- $\implies q$ has infinitely many stationary points along line $\mathbf{x}^* + \alpha \boldsymbol{v}_0$
- Since $\mathbf{H} = 2\mathbf{A}$, kind of stationary point not changing along $\boldsymbol{v}_0$